NAME

mbnavadjustmerge − Command line tool to create and modify MBnavadjust projects, including creating new projects, importing swath data, changing project settings, merging projects, adding global ties, setting tie modes, automatically picking crossing ties, checking for new crossings, exporting or importing crossing ties, and pre-generating the triangular meshes.

VERSION

Version 5.0

SYNOPSIS

mbnavadjustmerge --input=project-base
[--input=project_path
--output=project_path
--create-project=project_path
--section-length=km
--section-soundings=count
--contour-interval=meters
--color-interval=meters
--tick-interval=meters
--label-interval=meters
--decimation=value
--smoothing=value
--zoffsetwidth=meters
--import=path[:format]
--import-as-survey=path[:format]
--find-crossings
--autopick
--autopick-horizontal
--autopick-crossing-type=all|mediocre|good|better|true
--autopick-scope=all|survey|withsurvey|block|file|withfile|withsection
--autopick-survey=survey
--autopick-survey2=survey
--autopick-file=file
--autopick-section=section
--autopick-overlap-threshold=percent
--invert-navigation
--update-grids
--apply-navigation
--set-global-tie=file:section[:snav]/xoffset/yoffset/zoffset[/xsigma/ysigma/zsigma]
--set-global-tie-xyz=file:section[:snav]
--set-global-tie-xyonly=file:section[:snav]
--set-global-tie-zonly=file:section[:snav]
--set-all-global-ties-xyz
--set-all-global-ties-xyonly
--set-all-global-ties-zonly
--unset-global-tie=file:section
--unset-all-global-ties
--add-crossing=file1:section1/file2:section2
--set-tie=file1[:section1]/file2[:section2]/xoffset/yoffset/zoffset[/xsigma/ysigma/zsigma]
--set-tie=file1[:section1]/file2[:section2]///zoffset
--set-tie-xyz=file1:section1/file2:section2
--set-tie-xyonly=file1:section1/file2:section2

--set-tie-zonly=file1:section1/file2:section2
--set-ties-xyz-all
--set-ties-xyonly-all
--set-ties-zonly-all
--set-ties-xyz-with-file=file
--set-ties-xyonly-with-file=file
--set-ties-zonly-with-file=file
--set-ties-xyz-by-survey=survey
--set-ties-xyonly-by-survey=survey
--set-ties-zonly-by-survey=survey
--set-ties-xyz-by-block=survey1/survey2
--set-ties-xyonly-by-block=survey1/survey2
--set-ties-zonly-by-block=survey1/survey2
--set-ties-zoffset-by-block=survey1/survey2/zoffset
--set-ties-xyonly-by-time=timethreshold
--unset-tie=file1:section1/file2:section2
--unset-ties-with-file=file
--unset-ties-with-survey=survey
--unset-ties-by-survey=survey
--unset-ties-by-block=survey1/survey2
--unset-ties-all
--skip-unset-crossings
--unset-skipped-crossings-by-block=survey1/survey2
--unset-skipped-crossings-between-surveys
--insert-discontinuity=file:section
--remove-discontinuity=file:section
--merge-surveys=survey1:survey2
--update-file=file
--update-survey=survey
--update-all-files
--import-tie-list=file
--export-tie-list=file
--triangulate
--triangulate-all
--triangulate-scale=scale
--triangulate-section=file:section
--unset-short-section-ties=min_length
--skip-short-section-crossings=min_length
--verbose
--help]

DESCRIPTION

MBnavadjustmerge is a command line program that creates new **MBnavadjust** projects, imports swath data and changes project settings, automatically picks navigation ties at crossings, checks for new crossings, merges two existing **MBnavadjust** projects, or otherwise modifies a single **MBnavadjust** project. It provides command-line access to most of the project-management functionality found in the interactive **mbnavadjust** graphical program, making it possible to script and automate **MBnavadjust** project workflows.

MBnavadjust is an interactive graphical program used to adjust swath data navigation by matching bathymetric features in overlapping and crossing swaths. The primary purpose of **mbnavadjust** is to eliminate relative navigational errors in swath data obtained from poorly navigated sonars.

To create a new project, use the **--create-project** option, optionally combined with the section-length, section-soundings, contour-interval, color-interval, tick-interval, label-interval, decimation, smoothing, and zoffsetwidth settings options; these same settings options can be applied to an existing project as well. Swath data (a single file or a recursive datalist) can then be imported with the **--import** or **--import-as-survey** option. The **--find-crossings** option searches for overlapping and crossing swath sections among the imported data (this happens automatically as data are imported, so **--find-crossings** is mainly useful after a merge or after inserting/removing discontinuities). The **--autopick** and **--autopick-horizontal** options automatically pick navigation ties at unanalyzed crossings by computing and searching a bathymetric misfit grid, exactly as the interactive **Action->Auto-Pick Offsets** menu item does in **mbnavadjust**; the **--autopick-crossing-type**, **--autopick-scope**, **--autopick-survey**[**2**], **--autopick-file**, **--autopick-section**, and **--autopick-overlap-threshold** options narrow which crossings are considered, mirroring the crossing-type and scope filters available under the **View** menu in **mbnavadjust**.

Once ties have been picked, the **--invert-navigation** option solves the network of ties and crossings for a corrected navigation model, exactly as the interactive **Action->Invert Navigation** menu item does in **mbnavadjust**. The **--update-grids** option regenerates the project's reference bathymetry grids from the current navigation, equivalent to **Action->Update Grids**. The **--apply-navigation** option writes the corrected navigation solution out to the swath data files' processing parameters, equivalent to **Action->Apply Navigation**; this is normally the last step in a navigation adjustment workflow, run after **--invert-navigation**.

With respect to the merging function, if the **MBnavadjustmerge** user specifies two input projects and no output project, then the second project is added to the first. If an output project is specified, then the two projects are merged and the new combined project is output.

In order to make use of the project modification commands, a single input project must be specified, along with one or more of the modification commands. These can include creating a project, importing swath data, changing project settings, automatically picking ties, checking for new crossings, adding crossings, setting tie z-offset values, setting tie offsets (even if no corresponding crossing already exists), setting tie modes (xy only, z only, xyz), and deleting ties. When more than one modification command is given, they are applied in the order in which they appear on the command line (project settings changes are always applied first, since section splitting during import depends on the section-length and section-soundings settings).

MB-SYSTEM AUTHORSHIP

David W. Caress
Monterey Bay Aquarium Research Institute
Dale N. Chayes
Center for Coastal and Ocean Mapping
University of New Hampshire
Christian do Santos Ferreira
MARUM - Center for Marine Environmental Sciences
University of Bremen

OPTIONS

−-input=project-base

−-input=project-add The **--input** option defines an existing, input **MBnavadjust** project. A project may be defined by a relative or absolute path to the project *.nvh file or the project *.dir directory. This option can be used twice to specify two input projects. The first use of **--input** defines the base input project, and the second defines the add input project. In the case where the **--output** option is used to define an output project, the two input projects will both be copied to the new output project (*project-base* first followed by *project-add*). If the **output** option is not specified, then the *project-base* project is used as the output, and the *project-add* project is added on to the *project-base* project. If only the base input project is specified, then that project will be modified according to one or more of the "add", "delete", "skip", or "set" commands.

−-output=project-output

This option defines the new **MBnavadjust output project. The two input projects will both be copied to the new output project (***project-base* **first followed by** *project-add***). If the output option is not specified, then the** *project-base* **project is used as the output, and the** *project-add* **project is added on to the** *project-base* **project.**

−-create-project=project-path

This option creates a new, empty **MBnavadjust** project at *project-path*, exactly as the **File->New** menu item does in **mbnavadjust**. As with **--input**, a project may be defined by a relative or absolute path to the project *.nvh file or the project *.dir directory. This option cannot be combined with **--input**. Any of the section-length, section-soundings, contour-interval, color-interval, tick-interval, label-interval, decimation, smoothing, or zoffsetwidth settings options given on the same command line are applied to the new project as it is created; any not given default to the same values used by **mbnavadjust** for a new project. Other modification commands (e.g. **--import**, **--find-crossings**, **--autopick**) given on the same command line are then applied to the newly created project.

−-section-length=km

This option sets the maximum swath data section length in kilometers, exactly as the **Max Section Length (km)** slider does in the **mbnavadjust Controls** window. This setting only affects how files imported afterward (in the same command, or in a later command) are broken up into sections; it does not resection files already imported into the project.

−-section-soundings=count

This option sets the maximum number of soundings (not pings) that may be included in a single swath data section, exactly as the **Max # Soundings in Section** slider does in the **mbnavadjust Controls** window. Like **--section-length**, this only affects files imported afterward.

−-contour-interval=meters

This option sets the depth interval in meters for the bathymetric contours used in the **mbnavadjust Nav Err** window, exactly as the **Contour Interval (m)** slider does in the **Controls** window.

−-color-interval=meters

This option sets the depth interval in meters at which the color of the bathymetric contours changes, exactly as the **Color Interval (m)** slider does in the **mbnavadjust Controls** window. This should be a multiple of the contour interval.

−-tick-interval=meters

This option sets the depth interval in meters at which contours have downhill facing tickmarks, exactly as the **Tick Interval (m)** slider does in the **mbnavadjust Controls** window. This should be a multiple of the contour interval.

−-label-interval=meters

This option sets the depth interval in meters at which bathymetric contours are annotated with their depth value.

−-decimation=value

This option sets the ping decimation used when calculating bathymetric contours for the **mbnavadjust Nav Err** window, exactly as the **Decimation** slider does in the **Controls** window. A value of 1 uses all soundings; a value of 2 uses soundings from every second ping, and so on.

−-smoothing=value

This option sets the importance of smoothing in the adjusted navigation model calculated by **Action->Invert Navigation** in **mbnavadjust**, exactly as the **Inversion Smoothing** slider does in the **Controls** window. Larger values yield a smoother model, smaller values a rougher model; the default is 3.0 and the useful range is about 0.10 to 10.0.

−-zoffsetwidth=meters

This option sets the width in meters of the vertical offset search range used when computing the bathymetric misfit grid for a crossing (both interactively in the **mbnavadjust Nav Err** window and by

--autopick).

−-import=path[:format]

This option imports swath data into the project, exactly as the **File->Import Swath Data** menu item does in **mbnavadjust**. If *path* refers to a single swath data file, the *format* id must be given explicitly as *path***:***format*; unlike the graphical file selection dialog, the **MB-System** filename suffix convention is not used to infer the format here. If no *format* is given, *path* is instead read as a recursive **MB-System** datalist naming one or more swath files (see the **MB-System** manual page for a description of recursive datalists), and the format of each listed file is taken from the datalist. As with the interactive import, each file is broken into sections according to the current section-length and section-soundings settings, and any new crossings with previously imported data are found automatically. This option may be given more than once on the same command line to import multiple files or datalists in sequence. Swath files that continue in time from the file imported immediately before them are treated as part of the same survey; a time gap starts a new survey, unless **--import-as-survey** is used instead.

−-import-as-survey=path[:format]

This option behaves exactly like **--import**, except that every file imported by this option (including a multi-file datalist) is forced into a single new survey regardless of any time gaps between files, exactly as selecting the **Survey** file mode does in the **mbnavadjust** file import dialog.

−-find-crossings

This option searches for new crossings among the swath data sections already imported into the project, exactly as the **Action->Check For New Crossings** menu item does in **mbnavadjust**. The percentage of overlap is also recalculated for all existing crossings. This happens automatically as data are imported with **--import**, so **--find-crossings** is mainly useful after a **--merge-surveys**, after inserting or removing a discontinuity, or after merging two projects.

−-autopick

This option automatically picks navigation ties at crossings that have not yet been analyzed, exactly as the **Action->Auto-Pick Offsets** menu item does in **mbnavadjust**. For each qualifying crossing, the swath data are loaded, a three-dimensional (longitude, latitude, and vertical offset) bathymetric misfit grid is computed, and a tie is added at the offset with the smallest misfit *if* the uncertainty in that minimum is small relative to the size of the overlap region; otherwise the crossing is left unset. As with interactive auto-picking, unsupervised offset picks are frequently erroneous, so the resulting ties should be reviewed (e.g. in **mbnavadjust**, or with **--export-tie-list**) before relying on them for a navigation inversion. By default all unanalyzed crossings are considered; use **--autopick-crossing-type**, **--autopick-scope**, and the associated survey/file/section options to narrow the crossings considered, and **--autopick-overlap-threshold** to change the minimum overlap percentage required.

−-autopick-horizontal

This option behaves exactly like **--autopick**, except that only the horizontal (longitude and latitude) offset of the minimum misfit is used; the vertical offset is left unchanged. This corresponds to the horizontal-only auto-pick behavior available in the interactive **mbnavadjust** "Nav Err" window, and is appropriate when the data are already corrected for vertical offsets (e.g. tides).

−-autopick-crossing-type=all|mediocre|good|better|true

This option limits which crossings **--autopick** or **--autopick-horizontal** consider, mirroring the crossing-type items under the **View** menu in **mbnavadjust**: *all* considers any crossing meeting the overlap threshold (the default), *mediocre* requires at least 10% overlap, *good* at least 25% overlap, *better* at least 50% overlap, and *true* restricts to crossings where the navigation tracks actually cross ("true crossings").

−-autopick-scope=all|survey|withsurvey|block|file|withfile|withsection

This option limits which crossings **--autopick** or **--autopick-horizontal** consider by survey, file, or section, mirroring the **View->Show Only Selected ...** items in **mbnavadjust**: *all* considers every crossing (the default); *survey* restricts to crossings where both sections belong to the survey given by **--autopick-survey**; *withsurvey* restricts to crossings where either section belongs to that survey;

block restricts to crossings between the two surveys given by **--autopick-survey** and **--autopick-survey2**; *file* restricts to crossings where both sections belong to the file given by **--autopick-file**; *withfile* restricts to crossings where either section belongs to that file; and *withsection* restricts to crossings involving the specific section given by **--autopick-file** and **--autopick-section**.

−-autopick-survey=survey

This option sets the survey used to filter crossings when **--autopick-scope** is *survey*, *withsurvey*, or *block* (as the first of the two surveys in the *block* case).

−-autopick-survey2=survey

This option sets the second survey used to filter crossings when **--autopick-scope** is *block*.

−-autopick-file=file

This option sets the file used to filter crossings when **--autopick-scope** is *file*, *withfile*, or *withsection*.

−-autopick-section=section

This option sets the section (within the file given by **--autopick-file**) used to filter crossings when **--autopick-scope** is *withsection*.

−-autopick-overlap-threshold=percent

This option sets the minimum percentage of overlap a crossing must have to be considered by **--autopick** or **--autopick-horizontal**, overriding the default threshold of 10%. This is a floor applied in addition to whatever minimum overlap is implied by **--autopick-crossing-type**.

−-invert-navigation

This option solves the network of navigation ties and crossings for a corrected navigation model, exactly as the **Action->Invert Navigation** menu item does in **mbnavadjust**. The inversion is a sparse least-squares solve over all files and sections in the project using their current ties and global ties; it requires that ties already exist (e.g. from **--autopick** or **--set-tie**) and updates the project's navigation solution and per-section offsets in place. This does not modify the swath data files themselves; use **--apply-navigation** afterward to do so.

−-update-grids

This option regenerates the project's reference bathymetry grids from the current navigation solution, exactly as the **Action->Update Grids** menu item does in **mbnavadjust**. This is normally run after **--invert-navigation** so that the reference grids reflect the corrected navigation, and requires **mbdatalist**, **mbgrid**, and **mbprocess** to be available.

−-apply-navigation

This option writes the current navigation solution out to the swath data files' processing parameters, exactly as the **Action->Apply Navigation** menu item does in **mbnavadjust**. This is normally the last step in a navigation adjustment workflow, run after **--invert-navigation**; running it before an inversion has been computed for the project has no effect on the data files.

−-set-global-tie=file:section[:snav]/xoffset/yoffset/zoffset[/xsigma/ysigma/zsigma]

This options sets a global tie for a navigation point in the project. Global ties are ties to the global reference frame, which generally means GPS positions (e.g. WGS84). The file and section id's must be specified, along with the x (east-west), y (north-south), and z (vertical positive up) offset values in meters. If the snav id (navigation point id) is not specified, it is assumed to be 0, the first navigation point in the specified section. If the uncertainties in the global tie (xsigma, ysigma, zsigma) are not specified, then xsigma = ysigma = 10 m and zsigma = 0.5 m are assumed.

−-set-global-tie-xyz=file:section[:snav]

This option sets the mode of the existing specified global tie in the **MBnavadjust** project to be "XYZ". This means all three coordinates of the tie will be used as constraints in the inversion for an optimal navigation model.

−-set-global-tie-xyonly=file:section[:snav]

This option sets the mode of the existing specified global tie in the **MBnavadjust** project to be "XY". This means only the horizontal coordinates of the tie will be used as constraints in the inversion for an optimal navigation model.

−-set-global-tie-zonly=file:section[:snav]
This option sets the mode of the existing specified global tie in the **MBnavadjust** project to be "Z". This means only the vertical coordinate of the tie will be used as a constraint in the inversion for an optimal navigation model.

−-set-all-global-ties-xyz
This option sets the mode of all global ties in the **MBnavadjust** project to be "XYZ". This means all three coordinates of the tie will be used as constraints in the inversion for an optimal navigation model.

−-set-all-global-ties-xyonly
This option sets the mode of all global ties in the **MBnavadjust** project to be "XY". This means only the horizontal coordinates of the tie will be used as constraints in the inversion for an optimal navigation model.

−-set-all-global-ties-zonly
This option sets the mode of all global ties in the **MBnavadjust** project to be "Z". This means only the vertical coordinate of the tie will be used as a constraint in the inversion for an optimal navigation model.

−-unset-global-tie=file:section
This option unsets (deletes) the specified global tie in the **MBnavadjust** project.

−-add-crossing=file1:section1/file2:section2
This option adds the specified crossing to the **MBnavadjust** project.

−-set-tie=file1[:section1]/file2[:section2]/[xoffset]/[yoffset]/zoffset[/xsigma/ysigma/zsigma]
This option adds the specified tie to the **MBnavadjust** project. If the corresponding crossing does not already exist, it will be created. If a section is not specified, then it is assumed to be the first value, i.e. 0. If the tie offsets are specified but the uncertainties (xsigma, ysigma, zsigma) are not specified, then xsigma = ysigma = 10 m and zsigma = 0.5 m are assumed.

−-set-tie=file1[:section1]/file2[:section2]///zoffset
This option modifies the z offset value of the specified tie in the **MBnavadjust** project. If a section is not specified, then it is assumed to be the first value, i.e. 0. The specified tie must already exist, and this command changes the z offset value while leaving the x and y offsets and the uncertainties unchanged.

−-set-tie-xyz=file1:section1/file2:section2
This option sets the mode of the existing specified tie in the **MBnavadjust** project to be "XYZ". This means all three coordinates of the tie will be used as constraints in the inversion for an optimal navigation model.

−-set-tie-xyonly=file1:section1/file2:section2
This option sets the mode of the existing specified tie in the **MBnavadjust** project to be "XY". This means only the horizontal coordinates of the tie will be used as constraints in the inversion for an optimal navigation model.

−-set-tie-zonly=file1:section1/file2:section2
This option sets the mode of the existing specified tie in the **MBnavadjust** project to be "Z". This means only the vertical coordinate of the tie will be used as a constraint in the inversion for an optimal navigation model.

−-unset-tie=file1:section1/file2:section2
This option unsets (deletes) the specified tie in the **MBnavadjust** project.

−-set-ties-xyz-all
This option sets the mode of all ties in the **MBnavadjust** project to be "XYZ". This means all three coordinates of these ties will be used as constraints in the inversion for an optimal navigation model.

−-set-ties-xyonly-all
This option sets the mode of all ties in the **MBnavadjust** project to be "XY". This means only the horizontal coordinates of these ties will be used as constraints in the inversion for an optimal navigation model.

−-set-ties-zonly-all
This option sets the mode of all ties in the **MBnavadjust** project to be "Z". This means only the vertical coordinate of these ties will be used as constraints in the inversion for an optimal navigation model.

−-set-ties-xyz-with-file=file
This option sets the mode of all ties involved with the specified file in the **MBnavadjust** project to be "XYZ". This means all three coordinates of these ties will be used as constraints in the inversion for an optimal navigation model.

−-set-ties-xyonly-with-file=file
This option sets the mode of all ties involved with the specified file in the **MBnavadjust** project to be "XY". This means only the horizontal coordinates of these ties will be used as constraints in the inversion for an optimal navigation model.

−-set-ties-zonly-with-file=file
This option sets the mode of all ties involved with the specified file in the **MBnavadjust** project to be "Z". This means only the vertical coordinate of these ties will be used as constraints in the inversion for an optimal navigation model.

−-set-ties-xyz-with-survey=survey
This option sets the mode of all ties involved with the specified survey in the **MBnavadjust** project to be "XYZ". This means all three coordinates of these ties will be used as constraints in the inversion for an optimal navigation model.

−-set-ties-xyonly-with-survey=survey
This option sets the mode of all ties involved with the specified survey in the **MBnavadjust** project to be "XY". This means only the horizontal coordinates of these ties will be used as constraints in the inversion for an optimal navigation model.

−-set-ties-zonly-with-survey=survey
This option sets the mode of all ties involved with the specified survey in the **MBnavadjust** project to be "Z". This means only the vertical coordinate of these ties will be used as constraints in the inversion for an optimal navigation model.

−-set-ties-xyz-by-survey=survey
This option sets the mode of all ties between two sections in the specified survey in the **MBnavadjust** project to be "XYZ". This means all three coordinates of these ties will be used as constraints in the inversion for an optimal navigation model.

−-set-ties-xyonly-by-survey=survey
This option sets the mode of all ties between two sections in the specified survey in the **MBnavadjust** project to be "XY". This means only the horizontal coordinates of these ties will be used as constraints in the inversion for an optimal navigation model.

−-set-ties-zonly-by-survey=survey
This option sets the mode of all ties between two sections in the specified survey in the **MBnavadjust** project to be "Z". This means only the vertical coordinate of these ties will be used as constraints in the inversion for an optimal navigation model.

−-set-ties-xyz-by-block=survey1/survey2
This option sets the mode of all ties between sections of the specified two surveys in the **MBnavadjust** project to be "XYZ". This means all three coordinates of these ties will be used as constraints in the inversion for an optimal navigation model.

−-set-ties-xyonly-by-block=survey1/survey2
This option sets the mode of all ties between sections in the specified two surveys in the **MBnavadjust** project to be "XY". This means only the horizontal coordinates of these ties will be used as constraints in the inversion for an optimal navigation model.

−-set-ties-zonly-by-block=survey1/survey2
This option sets the mode of all ties between sections in the specified two surveys in the **MBnavadjust** project to be "Z". This means only the vertical coordinate of these ties will be used as constraints in the inversion for an optimal navigation model.

−-set-ties-zoffset-by-block=survey1/survey2/zoffset
This option resets the zoffset value of all ties between sections in the specified two surveys.

−-set-ties-xyonly-by-time=dt
This option sets the mode of all ties with nav points separated in time by dt or larger in the **MBnavadjust** project to be "XY". This means only the horizontal coordinates of these ties will be used as constraints in the inversion for an optimal navigation model.

−-unset-ties-with-file=file
This option unsets (deletes) all ties involved with the specified file in the **MBnavadjust** project.

−-unset-ties-with-survey=survey
This option unsets (deletes) all ties involved with the specified survey in the **MBnavadjust** project.

−-unset-ties-by-survey=survey
This option unsets (deletes) all ties between two sections in the specified survey in the **MBnavadjust** project.

−-unset-ties-by-block=survey1/survey2
This option unsets (deletes) all ties between sections of the specified two surveys in the **MBnavadjust** project.

−-unset-ties-all
This option unsets (deletes) all ties in the **MBnavadjust** project. The command −-unset-all-ties is also accepted.

−-skip-unset-crossings
This option sets all unset crossings in the input projects to be skipped. This does not apply to any new crossings between the two merged projects.

−-unset-skipped-crossings-by-block=survey1/survey2
This option sets all skipped crossings in the specified two surveys to be unset.

−-unset-skipped-crossings-between-surveys
This option sets all skipped crossings between different surveys in the input projects to be unset. This does not apply to any new crossings between the two merged projects.

−-insert-discontinuity=file:section
This option inserts a navigation discontinuity immediately after the specified file and section.

−-remove-discontinuity=file:section
This option removes a navigation discontinuity immediately after the specified file and section (if the discontinuity exists).

−-merge-surveys=survey1:survey2
This option merges two surveys.

−-update-file=file
This option updates the beam flags (which soundings are valid and which are ignored) for the specified file's sections to match the current processed version of that swath file. Pings in each section are matched to pings in the current processed file by timestamp, and only the beam-flag byte is changed; the section's navigation, attitude, bathymetry, amplitude, and sidescan values are all preserved

unchanged. This allows bad soundings removed by re-editing (e.g. with **mbclean** or **mbedit**) after the file was originally imported to be excluded from the mbnavadjust project as well. If a matched ping's beam count differs from what is stored in the project (e.g. because the file has been reprocessed with a different sonar model or binning), that ping's flags are left unchanged and a warning is printed. The coverage mask and triangulated surface for every updated section are automatically recomputed to reflect the new flags.

−-update-survey=survey
This option updates the beam flags for every file in the specified survey (block), as described above for **−-update-file**.

−-update-all-files
This option updates the beam flags for every file in the project, as described above for **−-update-file**.

−-import-tie-list=filename
Import a listing of navigation ties that has been exported from a different **mbnavadjust** project using the **--export-tie-list** command.

−-export-tie-list=filename
Output the current navigation ties as a text file that can be imported into a different **mbnavadjust** project using the **--import-tie-list** command. The ties that are output reflect all modifications specified by other commands.

−-triangulate
This command causes **mbnavadjustmerge** to pre-generate triangular meshes used by **mbnavadjust** for contouring each section as part of the graphical analysis of crossings. These meshes are stored within the project directory as a *.tri file in parallel with each *.mb71 section file. Any section triangle mesh files that already exist are not recreated. The region including each section is divided into a grid with a cell size determined by the *scale* value set by the **−-triangulate-scale** option; the triangle vertices are selected as the least deep (shoalest) soundings within each grid cell.

−-triangulate-all
This command causes **mbnavadjustmerge** to pre-generate triangular meshes used by **mbnavadjust** for contouring each section as part of the graphical analysis of crossings. All section triangle mesh files are created; any pre-existing triangle files are overwritten.

−-triangulate-scale=scale
This option sets the scale (size) of the triangles generated by the **−-triangulate** or **−-triangulate-all** commands. By default, the scale is calculated as 1/100 of the width or height (whichever is longer) of the region covered by the section.

−-triangulate-section=file:section
This command causes **mbnavadjustmerge** to pre-generate the triangular mesh for the specified section.

−-unset-short-section-ties=min_length
This option unsets (deletes) all ties for which one or both data sections have a navigation track length less than *min_length*.

−-skip-short-section-crossings=min_length
This option sets all crossings for which one or both data sections have a navigation track length less than *min_length* to "skipped" mode. Any ties associated with these crossings are deleted.

−-remake-mb166-files
This option causes **mbnavadjustmerge** to remake the navigation files inside the project, fixing problems caused by a since-fixed bug.

−-fix-sensordepth
This option causes **mbnavadjustmerge** to recalculate the sensordepth values in the project file from the imported swath data, fixing problems caused by a since-fixed bug.

−-verbose
This option increases the verbosity of **MBnavadjustmerge**, which means that more information than by default is output to the stderr stream of the shell.

−-help
This option causes **MBnavadjustmerge** to output a list of the possible command line options, and then exit.

EXAMPLES

Suppose you want to build a new **MBnavadjust** project from a survey's recursive datalist and automatically pick as many navigation ties as possible without ever opening the interactive **mbnavadjust** program. The following creates the project, imports the datalist, and auto-picks ties at every unanalyzed crossing:
mbnavadjustmerge --create-project=Navadjust20240510.nvh --section-length=0.1 .br
mbnavadjustmerge --input=Navadjust20240510.nvh --import=datalist.mb-1 .br
mbnavadjustmerge --input=Navadjust20240510.nvh --autopick --autopick-scope=all
The resulting project can then be opened in **mbnavadjust** to review the automatically picked ties before running **Action->Invert Navigation**, or the ties can be reviewed non-interactively with **--export-tie-list**. Since unsupervised offset picks are frequently erroneous, reviewing them is recommended.

Suppose you have two AUV survey missions, 20140515m1 and 10140515m2, that overlap slightly. If you have used **MBnavadjust** to adjust the navigation of the two missions separately, you can use **MBnavadjustmerge** to merge the two **MBnavadjust** projects into a single new project without losing any of the ties made between overlapping sections in the existing projects. If the two existing projects are named "Navadjust20140515m1" and "Navadjust20140515m2", respectively, then there exist project files with a ".nvh" suffix and project directories with a ".dir" suffix. To create a new **MBnavadjust** project combining the two existing projects, the following will suffice:
mbnavadjustmerge --input=Navadjust20140515m1.nvh --input=Navadjust20140515m2.nvh --output=Navadjust20140515All.nvh

The new project Navadjust20140515All can be opened and analyzed further using **MBnavadjust**. When opening the new project, the user should first solve for a comprehensive navigation adjustment model by selecting the <Action->Invert Navigation> menu item, and then find the crossings between the two previously separate missions by selecting the <Action->Check For New Crossings> menu item.

If the **--skip-unset-crossings** option is added to the above command, then all unset crossings in the two input projects will be set to "skipped" mode in the output project.

SEE ALSO

mbsystem(1), **mbio**(1), **mbprocess**(1), **mbnavadjust**(1), **mbset**(1)

BUGS

It started out simple and bulletproof, but now it's too complicated to be bulletproof. It's probably nerfgunproof, though. Good luck.